

MRI-Guided Microwave Thermal Ablation

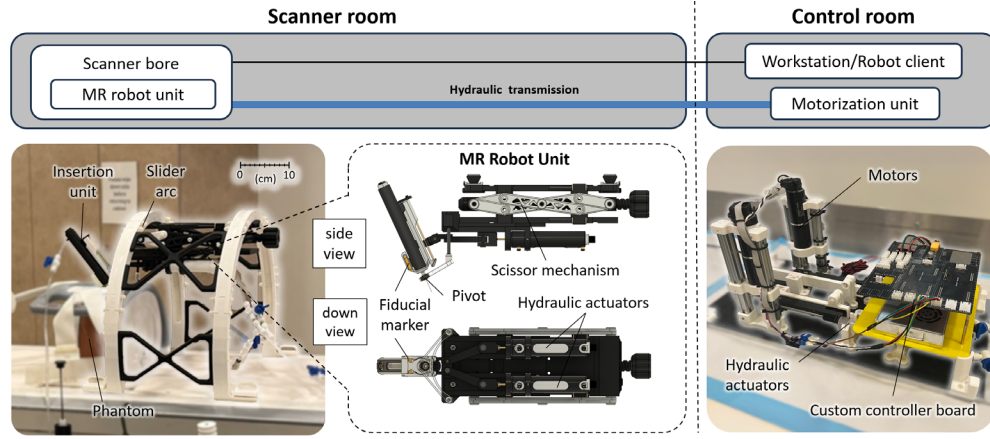
Formulation as a PDE-Constrained Optimal Control Problem

Free Final Time, State-Constrained Variational Formulation

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270C Final Project Proposal

MRI Surgical Robot Hardware



1. Problem Overview and Physical Setting

MRI-guided thermal ablation is the clinical procedure of using a controlled energy source (microwave, radiofrequency, laser, or focused ultrasound) to heat and destroy a tumor target while sparing surrounding healthy tissue. Real-time MRI temperature imaging (MR thermometry, via the proton resonance frequency shift) provides online feedback of the three-dimensional temperature field, making this a natural setting for closed-loop, feedback-optimal control.

General surgical procedures are shown as follows:

1. Identify tumor area using 3D MRI volume scan
2. Locate insertion point and depth from surgeon's experience
3. Use robotics system to align ablation probe to the target point
4. Start ablation with MRI thermometry feedback
5. Check ablation zone with volume scan
6. Retract probe and finish

The goal is to create an ablation zone just big enough to cover the tumor area with roughly 1-cm boarder and prevent killing surrounding healthy tissue.

1.1 Spatial Coordinates

We denote the tissue domain by $\Omega \subset \mathbb{R}^3$ (or the cylindrical domain $\Omega \subset \mathbb{R}^2$ in the axisymmetric (r, z) case). The boundary is $\partial\Omega$, with inward unit normal $\hat{n}$. The time horizon is $t \in [0, t_f]$, where the final time $t_f > 0$ is a free optimization variable.

Cartesian: $r = (x, y, z) \in \Omega \subset \mathbb{R}^3$

Cylindrical (axisymmetric): $r = (r, \theta, z)$

MRI voxel index (discretized): $i \in \{1, 2, \dots, N\}$, $N = N_x \times N_y \times N_z$

2. State Variables

The system state is a pair of spatially-distributed fields:

2.1 Temperature Field

$$T = T(r, t): \Omega \times [0, t_f] \rightarrow \mathbb{R}[\text{°C or K}]$$

$T(r, t)$ is the primary observable — it is measured in real time by MRI thermometry. In the discretized system, the temperature state vector is:

$$x_T(t) = [T_1(t), T_2(t), \dots, T_N(t)]^T \in \mathbb{R}^N$$

where $T_i(t) = T(r_i, t)$ is the temperature at the i -th voxel centroid r_i

2.2 Thermal Damage Field (Arrhenius)

$$\Omega_d = \Omega_d(r, t): \Omega \times [0, t_f] \rightarrow \mathbb{R}^+[\text{dimensionless}]$$

Ω_d accumulates irreversible tissue damage. It is defined by the Arrhenius thermal dose model:

$$\Omega_d(r, t) = \int_0^t A \cdot \exp\left(-E_a/(R \cdot T(r, \tau))\right) d\tau$$

Equivalently in ODE form:

$$\frac{d\Omega_d}{dt} = A \cdot e^{-E_a/(R \cdot T)}, \Omega_d(r, 0) = 0$$

where

A [**s⁻¹**]: frequency factor

E_a [**J/mol**]: activation energy

$R = 8.314$ [**J/(mol·K)**]: universal gas constant

The threshold $\Omega_d = 1$ corresponds to **63%** probability of cell death (irreversible necrosis).

2.3 Combined Distributed State

The complete infinite-dimensional state is:

$$\mathcal{X}(r, t) = (T(r, t), \Omega_d(r, t)), r \in \Omega, t \in [0, t_f]$$

After spatial discretization, the lumped finite-dimensional state vector is:

$$x(t) = [x_T(t)^T, x_{\Omega}(t)^T]^T \in \mathbb{R}^{2N}$$

3. Control Variables

The control input depends on the ablation modality. We consider the general microwave/RF case:

3.1 Scalar Power Control (Single Applicator)

$$u(t) = P(t) \in [0, P_{max}][W]$$

3.2 Vector Control (Robotic MRI-Guided Ablation)

For a robotic system with actuated probe position and orientation:

$$u(t) = [P(t), q(t)]^T \in \mathbb{R}^{(1+n_q)}$$

where $q(t) \in \mathbb{R}^{n_q}$ is the robot joint configuration vector.

3.3 Control Constraints

$$u(t) \in \mathcal{U} := u: 0 \leq P(t) \leq P_{max}, |\dot{u}(t)| \leq \dot{u}_{max}$$

The control set $\mathcal{U}$ encodes physical actuator limits (maximum power, maximum slew rate, probe mechanical limits).

4. System Dynamics (State Equations)

4.1 Pennes Bioheat Equation (Continuous PDE)

The temperature field is governed by the Pennes bioheat equation:

$$\rho(r)c(r)\frac{\partial T}{\partial t} = \nabla \cdot (k(r)\nabla T) - \omega_b(r)\rho_b c_b (T - T_b) + Q_{met}(r) + Q_{source}(r, u, t)$$

where the terms are defined in the following table:

- $\rho(r)$ [**kg/m³**]: tissue mass density
- $c(r)$ [**J/(kg·K)**]: tissue specific heat capacity
- $k(r)$ [**W/(m·K)**]: thermal conductivity (possibly anisotropic tensor)
- $\omega_b(r)$ [**s⁻¹**]: blood perfusion rate
- $\rho_b = 1050$ [**kg/m³**]: blood density
- $c_b = 3600$ [**J/(kg·K)**]: blood specific heat
- $T_b = 37$ [**°C**]: arterial blood temperature (body core, ~37°C)
- $Q_{met}(r)$ [**W/m³**]: metabolic heat generation rate
- $Q_{source}(r, u, t)$ [**W/m³**]: volumetric heat deposition from ablation source

4.2 Microwave Heat Source Model

For microwave ablation, the heat source is proportional to the Specific Absorption Rate (SAR):

$$Q_{source}(r, u, t) = SAR(r, q(t)) \cdot P(t) = \sigma(r)|E(r, q(t))|^2 \cdot P(t)$$

where $\sigma(r)$ [**S/m**] is tissue electrical conductivity and $E(r, q(t))$ [**V/m**] is the normalized electric field pattern (precomputed from a finite element EM model). The spatial pattern $b(r, q(t))$ encodes the antenna near-field distribution:

$$Q_{source}(r, u, t) = b(r, q) \cdot P(t)$$

4.3 Damage Dynamics

$$\frac{\partial \Omega_d}{\partial t} = A \cdot e^{-E_a/(R \cdot T(r, t))}, r \in \Omega, t \in (0, t_f]$$

4.4 Discretized State-Space Form

Applying finite difference or finite element spatial discretization (mesh size h , N voxels) yields the ODE system:

$$\dot{x}(t) = f(x(t), u(t)) = A_d x_T(t) + B(u(t)) + d(x_T(t))$$

where:

- $A_d \in \mathbb{R}^{N \times N}$: sparse diffusion-perfusion matrix (tri- or penta-diagonal for FD, or FEM stiffness/mass matrix)
- $B(u(t)) \in \mathbb{R}^N$: control input vector, $B_i = b(r_i, q) \cdot P$
- $d(x_T(t)) \in \mathbb{R}^N$: Arrhenius damage rate vector, $d_i = A \cdot e^{-E_a/(R \cdot T(r,t))}$

Explicitly:

$$\dot{x}_T = \frac{1}{\rho c} \cdot [K_d x_T - W_b (x_T - T_b 1) + Q_m e t + B_p P(t)]$$

$$\dot{x}_\Omega = A \cdot e^{-E_a/(R \cdot x_T)}$$

[elementwise]

where $K_d \in \mathbb{R}^{N \times N}$ is the (FD or FEM) discrete Laplacian weighted by $k(r)$, and

$$W_b = \text{diag}(\omega_b, i \rho_b c_b)$$

5. Boundary and Initial Conditions

5.1 Initial Conditions

$$T(r, 0) = T_0(r), r \in \Omega[\text{pre-treatment baseline temperature}]$$

$$\Omega_d(r, 0) = 0, r \in \Omega[\text{no pre-existing thermal damage}]$$

Typically $T_0(r) \approx 37^\circ\text{C}$ (body temperature) throughout the domain, or can be the measured pre-treatment MRI thermometry map.

5.2 Boundary Conditions on $\partial\Omega$

Three standard options (select based on anatomy):

Dirichlet (Fixed Temperature — large vessel or body surface):

$$T(r, t) = T_\infty, r \in \partial\Omega_D$$

Neumann (Insulated — symmetry plane):

$$k(r) \frac{\partial T}{\partial n} = 0, r \in \partial\Omega_N$$

Robin (Convective Cooling — tissue-air or tissue-fluid interface):

$$k(r) \frac{\partial T}{\partial n} + h_c(T - T_\infty) = 0, r \in \partial\Omega_R$$

where

h_c [W/(m²·K)]: convective heat transfer coefficient

T_∞ [°C or K]: ambient temperature

6. Domain Partition

The tissue domain Ω is partitioned into three disjoint non-overlapping regions:

$$\Omega = \Omega_T \cup \Omega_H \cup \Omega_M, \Omega_T \cap \Omega_H = \Omega_T \cap \Omega_M = \Omega_H \cap \Omega_M = \emptyset$$

- $\Omega_T \subset \Omega$: tumor (target) region — to be fully ablated
- $\Omega_H \subset \Omega$: healthy tissue — to be protected
- $\Omega_M \subset \Omega$: safety margin / transition zone (optional)

These regions are defined from pre-treatment MRI segmentation.

7. Optimal Control Problem Formulation

7.1 Standard Bolza Form

The complete optimal control problem is stated in Bolza form:

$$\min_{u(\cdot), t_f} J[u, t_f] = \Phi(x(t_f), t_f) + \int_0^{t_f} L(x(t), u(t), t) dt$$

subject to the state dynamics, boundary conditions, control constraints, and terminal constraints detailed below.

7.2 Running Cost L (Lagrangian)

The running cost captures three competing objectives during the treatment:

$$L(x, u, t) = \alpha_1 \|u(t)\|^2 + \alpha_2 \int_{\Omega_H} \max(0, T(r, t) - T_{safe})^2 dr + \alpha_3$$

- $\alpha_1 \|u(t)\|^2$: penalize excessive energy deposition (reduce tissue damage, device wear)
- $\alpha_2 \int_{\Omega_H} \max(0, T(r, t) - T_{safe})^2 dr$: quadratic penalty for healthy tissue temperature exceeding threshold $T_{safe} \approx 42 - 45$ °C
- $\alpha_3 > 0$: constant rate cost on time; encourages minimum-time treatment

7.3 Terminal Cost Φ (Mayer Term)

The Mayer term penalizes incomplete tumor ablation at the final time t_f :

$$\Phi(x(t_f), t_f) = \gamma_1 \int_{\Omega_T} \max(0, 1 - \Omega_d(r, t_f))^2 dr + \gamma_2 t_f$$

- $\gamma_1 \int_{\Omega_T} \max(0, 1 - \Omega_d(r, t_f))^2 dr$: soft penalty for unablated tumor voxels
- $\gamma_2 t_f$: direct penalization of total procedure time

7.4 Complete Cost Functional

Combining the Lagrangian and Mayer terms:

$$\begin{aligned} J[u, t_f] = & \gamma_1 \int_{\Omega_T} \max(0, 1 - \Omega_d(r, t_f))^2 dr + \gamma_2 t_f \\ & + \int_0^{t_f} \left[\alpha_1 \|u(t)\|^2 + \alpha_2 \int_{\Omega_H} \max(0, T(r, t) - T_{safe})^2 dr + \alpha_3 \right] dt \end{aligned}$$

The weights $\alpha_1, \alpha_2, \alpha_3, \gamma_1, \gamma_2 > 0$ are tuning parameters reflecting clinical priorities.

8. Constraints

8.1 State Dynamics (Equality Constraint)

$$\dot{x}(t) = f(x(t), u(t)), t \in (0, t_f], x(0) = x^0 = [T_0^T, 0^T]^T$$

8.2 Path (State-Control) Inequality Constraints

These must be held for all times $t \in (0, t_f]$ and all spatial locations:

(a) Healthy tissue temperature limit:

$$T(r, t) \leq T_{safe}, \forall r \in \Omega_H, \forall t \in [0, t_f]$$

(b) Absolute safety temperature (critical structures):

$$T(r, t) \leq T_{crit}, \forall r \in \Omega_{crit} \subset \Omega_H, \forall t \in [0, t_f]$$

(c) Control box constraints:

$$0 \leq P(t) \leq P_{max}, \forall t \in [0, t_f]$$

(d) Control rate constraints (actuator slew limit):

$$\left| \frac{dP}{dt} \right| \leq \dot{P}_{max}, \forall t \in [0, t_f]$$

8.3 Terminal (Endpoint) Constraints

(a) Tumor ablation requirement (hard constraint or soft via Φ):

$$\Omega_d(r, t_f) \geq 1, \forall r \in \Omega_T$$

(b) Free final time positivity:

$$t_f > 0$$

(c) Maximum allowable treatment time:

$$t_f \leq T_{max}$$

8.4 Summary: Complete Constrained Problem

$$\begin{aligned} & \min_{u(\cdot) \in \mathcal{U}, t_f \in (0, T_{max}]} J[u, t_f] \text{ subject to:} \\ & \dot{x} = f(x, u), x(0) = x^0 \quad [dynamics] \\ & T(r, t) \leq T_{safe}, \forall r \in \Omega_H, t \in [0, t_f] \quad [path safety] \\ & 0 \leq P(t) \leq P_{max} \quad [control bounds] \\ & \Omega_d(r, t_f) \geq 1, \forall r \in \Omega_T \quad [terminal ablation] \end{aligned}$$

9. Parameter Summary

Table 1 below summarizes key physical parameters and their typical values for soft tissue ablation:

- ρ : tissue density — 1000–1100 kg/m³
- c : specific heat — 3400–3800 J/(kg·K)
- k : thermal conductivity — 0.4–0.6 W/(m·K)
- ω_b : blood perfusion rate — 0.003–0.01 s⁻¹
- T_b : blood temperature — 37°C
- A (Arrhenius): frequency factor — 3.1×10^{98} s⁻¹ (for protein denaturation)
- E_a (Arrhenius): activation energy — 6.28×10^5 J/mol
- R : universal gas constant — 8.314 J/(mol·K)
- T_{safe} : healthy tissue safety threshold — 42–45°C
- T_{crit} : critical structure absolute limit — 40°C
- P_{max} : maximum applicator power — 30–100 W
- T_{max} : maximum allowed procedure time — 10–30 min

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